PROJECT REPORT

On

HEART DISEASE PREDICTION USING MACHINE LEARNING

Submitted in Partial Fulfillment of Award of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

By

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ALLIANCE UNIVERSITY

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project work entitled "**HEART DISEASE PREDICTION USING MACHINE LEARNING** " submitted **by** C H RAKESH in partial fulfillment for the award of the degree of Bachelor of Technology degree of Bachelor of Technology (CSE) of Alliance University, is a bonafide work accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr R C Karpagalakshmi
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DECLARATION

I/We hereby declare that the project entitled "**HEART DISEASE PREDICTION USING MACHINE LEARNING**"submitted by me/us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (CSE) of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of Dr RC Karpagalakshmi

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

Name of the Student	University Registration Number	Signature
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1. INTRODUCTION

Heart disease is one of the leading causes of death worldwide. Early prediction plays a crucial role in reducing risks and saving lives.

Traditional diagnosis methods depend on medical expertise and tests, which may be time-consuming and costly. With the advancement of **Artificial Intelligence (AI)** and **Machine Learning (ML)**, it is possible to analyze patient data and predict diseases efficiently.

In this project, we use **Machine Learning techniques** to predict whether a person has heart disease based on medical parameters such as:

- Age
- Cholesterol level
- Blood pressure
- Maximum heart rate

The problem is treated as a **binary classification problem**:

- **0 → No Heart Disease**
- **1 → Heart Disease**

2. LITERATURE SURVEY

Traditional Methods

- Clinical diagnosis
- ECG analysis
- Blood tests

Limitations:

- Time-consuming
- Requires expert interpretation

Machine Learning Approaches

- Logistic Regression
- Decision Trees
- Random Forest
- Neural Networks

Key Findings

- Cholesterol and blood pressure are major indicators
- Data preprocessing improves accuracy
- Logistic Regression is simple and effective

3. SYSTEM DESIGN

Functional Modules

1. **Data Collection**
    - Dataset containing patient health records
2. **Data Preprocessing**
    - Handle missing values
    - Normalize data
3. **Feature Engineering**
    - Create new features like age × cholesterol
4. **Model Training**
    - Logistic Regression algorithm
5. **Prediction**
    - Predict heart disease (0 or 1)

4. SYSTEM IMPLEMENTATION

Programming Tools

- Python

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, r2_score

# Load dataset
df = pd.read_csv("/content/heart_disease_data.csv")

print("First 5 rows:")
print(df.head())

print("\nLast 5 rows:")
```

```
print(df.tail())

# Feature Engineering
df["age_chol"] = df["age"] * df["chol"]

# Correlation Heatmap
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()

# Scatter Plots
plt.scatter(df['age'], df['target'])
plt.xlabel("Age")
plt.ylabel("Heart Disease")
plt.title("Age vs Heart Disease")
plt.show()

plt.scatter(df['chol'], df['target'])
plt.xlabel("Cholesterol")
plt.ylabel("Heart Disease")
plt.title("Cholesterol vs Heart Disease")
plt.show()
```

```
# Model Training
X = df[['age','chol','thalach','trestbps']]
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LogisticRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

5. RESULTS AND DISCUSSION

Performance Metrics

- Accuracy
- Precision
- Recall
- Confusion Matrix

Observations

- High cholesterol → higher risk
- Higher age → increased probability
- Maximum heart rate affects prediction

Graph Analysis

- Scatter plots show relationship between features
- Heatmap shows correlation

Discussion

- Model performs well
- Accuracy is good for small dataset
- More data can improve results

6. TESTING

Testing Methods

1. **Train-Test Split**
    - 80% Training
    - 20% Testing
2. **Confusion Matrix**
    - True Positive
    - False Positive
    - True Negative
    - False Negative

Testing Goals

- Validate accuracy
- Avoid overfitting
- Ensure reliability

7. CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion

This project demonstrates that **Machine Learning can effectively predict heart disease**. Logistic Regression provides good accuracy and is easy to implement.

The system:

- Predicts heart disease efficiently
- Helps in early diagnosis
- Supports medical decision-making

Limitations

- Limited dataset
- Few features
- No real-time data

Future Enhancements

- Use Deep Learning models
- Add real-time patient data
- Develop mobile/web app
- Improve accuracy with large datasets
- Integrate with hospital systems

8. REFERENCES

- Scikit-learn Documentation
- Machine Learning Books
- Research Papers on Heart Disease Prediction

- Medical datasets (UCI Repository)